

PITTSFIELD-PENSTONE, IL, USA

WMO: 744663

Lat: 39.639N

Lon: 90.778W

Elev: 710

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.0	(c) 8.5	(d) -6.4	(e) 4.0	(f) 4.8	(g) -1.9	(h) 5.1	(i) 10.1	(j) 24.3	(k) 28.2	(l) 21.9	(m) 31.8	(n) 5.9	(o) 290	(p) 0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.2	(c) 93.1	(d) 75.6	(e) 90.6	(f) 75.5	(g) 88.5	(h) 74.4	(i) 79.8	(j) 88.0	(k) 78.1	(l) 86.7	(m) 76.8	(n) 85.3	(o) 6.7	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	146.4	84.5	75.4	136.6	83.1	73.8	129.5	81.5	43.9	88.2	42.1	87.1	40.6	85.5	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.7	(b) 18.4	(c) 16.5	(e) -4.9	(f) 97.7	(g) 5.1	(h) 3.8	(i) -8.6	(j) 100.4	(k) -11.6	(l) 102.6	(m) -14.4	(n) 104.7	(o) -18.1	(p) 107.5		
(4) 20.7	(5) 18.4	(6) 16.5	(7) -5.2	(8) 82.3	(9) 5.1	(10) 1.5	(11) -8.9	(12) 83.4	(13) -11.9	(14) 84.3	(15) -14.7	(16) 85.2	(17) -18.5	(18) 86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	28.5	31.0	43.9	55.0	65.1	73.0	76.4	75.4	68.9	55.5	44.1	33.4
	DBStd	19.33	12.07	11.74	11.38	9.64	8.64	6.00	5.30	5.48	7.47	9.67	9.83	10.06
	HDD50	2329	669	540	259	56	4	0	0	0	0	53	228	520
	HDD65	5160	1130	952	659	322	111	9	1	1	43	324	628	980
	CDD50	3904	4	9	69	205	473	691	820	788	566	223	51	5
	CDD65	1259	0	0	4	21	115	250	356	324	159	29	1	0
	CDH74	11831	0	0	29	233	921	2225	3616	3037	1503	257	10	0
	CDH80	4402	0	0	1	47	246	783	1477	1227	556	64	1	0
Wind	WSAvg	6.9	8.4	8.2	8.6	8.7	7.1	5.8	4.4	4.1	5.0	6.6	7.9	7.9
Precipitation	PrecAvg	36.8	1.6	1.5	3.1	3.5	4.1	3.8	4.1	3.2	3.6	3.0	2.7	2.3
	PrecMax	52.2	5.2	3.4	7.2	7.8	10.6	8.0	16.2	8.1	10.3	7.2	9.5	7.5
	PrecMin	23.7	0.1	0.0	0.6	0.6	0.4	1.3	0.3	0.8	0.2	0.4	0.5	0.2
	PrecStd	7.5	1.2	0.8	1.5	1.9	2.0	2.0	3.2	1.8	2.2	1.6	1.9	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.4	67.7	78.1	85.3	89.3	92.9	98.2	98.7	93.7	86.5	74.7	64.5
		MCWB	55.9	55.9	62.3	63.9	72.2	74.6	75.1	75.2	72.2	70.7	60.3	59.0
	2%	DB	55.6	60.7	72.2	80.7	85.7	90.2	92.7	93.3	90.2	81.0	69.5	58.2
		MCWB	50.1	52.8	59.6	61.5	70.0	75.0	77.7	75.2	73.1	65.8	58.9	53.6
	5%	DB	50.5	54.5	66.3	75.6	82.4	88.0	90.3	90.3	87.1	75.5	64.0	52.8
		MCWB	44.8	48.0	57.0	60.1	67.9	74.4	76.8	75.0	72.1	63.1	55.9	48.0
	10%	DB	45.2	48.3	61.5	72.0	79.5	84.5	88.1	87.5	82.3	71.8	59.6	47.9
		MCWB	40.9	43.0	53.5	58.6	66.4	72.7	76.0	73.4	68.8	62.2	52.1	44.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	59.9	65.3	67.5	75.4	79.2	82.4	81.9	78.2	73.5	64.4	60.7
		MCDB	62.2	64.9	75.0	78.9	84.8	87.4	89.8	89.5	87.3	82.1	70.1	64.0
	2%	WB	51.6	54.1	61.7	65.2	72.9	77.5	80.3	79.2	76.2	70.7	60.5	54.7
		MCDB	54.8	58.0	69.2	74.7	82.0	86.8	88.4	87.6	85.1	77.6	66.2	57.3
	5%	WB	45.8	48.3	59.0	63.1	71.0	75.9	78.6	77.4	74.3	67.5	56.9	48.5
		MCDB	49.6	54.4	65.4	72.4	79.1	84.8	87.0	86.1	82.4	72.3	62.7	51.6
	10%	WB	40.5	42.9	54.7	60.3	68.9	74.3	77.2	75.9	72.4	63.9	53.5	43.9
		MCDB	45.5	48.4	60.7	69.3	76.7	82.7	85.6	83.6	79.5	68.7	59.0	47.4
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.3	18.2	20.2	18.6	18.3	19.2	19.4	21.7	19.9	17.5	14.2
		MCDBR	22.3	24.7	23.5	26.0	21.2	20.7	21.0	23.0	24.3	25.3	23.6	21.6
		MCWBR	17.5	17.6	13.7	13.0	9.8	9.5	9.3	9.2	9.8	12.6	15.2	16.9
	5% WB	MCDBR	20.5	23.6	21.2	21.5	18.7	18.9	18.7	18.7	20.3	19.7	21.1	20.9
		MCWBR	17.5	18.8	13.5	13.4	10.0	9.8	10.1	9.5	9.4	12.9	16.1	18.2
Clear-Sky Solar Irradiance	taub	0.296	0.315	0.338	0.389	0.432	0.431	0.457	0.447	0.400	0.351	0.304	0.290	
	taud	2.459	2.393	2.415	2.266	2.211	2.258	2.175	2.213	2.305	2.446	2.522	2.495	
	Ebn at Noon	278	287	292	281	270	270	261	260	266	270	274	272	
	Edh at Noon	25	31	34	42	45	43	46	43	37	29	23	23	
All-Sky Solar Radiation	RadAvg	620	879	1183	1545	1758	2000	1991	1749	1511	1043	704	525	
	RadStd	70	84	90	105	136	172	124	101	96	116	64	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+140

Nomenclature: See separate page